



The Hypothesis Testing Framework Transcript

Hello everyone, welcome to lesson 2 video 1. In this video, we will build the framework for hypothesis testing. This framework provides a structured way to test claims or assumptions about population parameters using sample data. Hello and welcome, in this session we are going to demystify one of the most powerful tools in statistics, which is called as the hypothesis testing.

This is the formal process that scientists, engineers, and analysts use to make decisions based on data. It allows us to move beyond simple observations and start drawing statistically sound conclusions about claims we encounter in everyday life. By the end of this lecture, you will understand the logic, the steps, and the concepts behind essential framework of hypothesis testing.

Okay, so let's begin. To understand the logic of hypothesis testing, you don't need to be a statistician, you just need to understand the principles of the analogy we are going to derive here, which is courtroom trial. You might have seen all of this in movies or you might have already witnessed how courtroom trial happens, right? This entire analogy helps us have the entire intuition of null hypothesis, alternate hypothesis, or the entire framework of hypothesis very well, to understand it very well.

In a fair trial, we have a default assumption, so on the left hand side you have the null hypothesis, right? So when we have a fair trial, we have a default assumption, the defendant is innocent until proven guilty. This is our starting point, our status quo. In statistics, we call the null hypothesis or H_0 , this is our H_0 , right? We say that the defendant is innocent until the defendant is proven guilty and this is what we call it as null hypothesis.

The alternate hypothesis, on the other hand, we have the prosecutor, right? The prosecutor presents a new claim, the defendant is guilty. So on the other hand, prosecutor is willing to make the defendant guilty. This is the alternate hypothesis.

This is the theory we are trying to find evidence for. So in general sense, what we do is, once we have all our sample collected and everything we have got, we try to frame hypothesis. Hypothesis has two parts, the null hypothesis and alternate hypothesis.

In null hypothesis, we will try to keep everything that we know so far. In alternate hypothesis, we will try to keep everything that we would want to prove because once there is a criminal, we would want the criminal to be proven criminal, right? That's what the prosecutor is doing in alternate hypothesis. So let's say if you are willing to find out the average battery life of iPhones is less than 12 hours, okay? So your alternate hypothesis, just put that assumption that you are making, is less than 12 hours.



In your null hypothesis, you will say that it is equals to 12 hours, right? On your alternate hypothesis, you will say that it is less than 12 hours. So that's what the entire hypothesis is because this is also required because you are majorly in situations wherein you would need answers from the sample. As what you are looking in the sample, let's say for example, the average marks of students in maths board exam is 85.

Now in your school, you are working as a maths teacher. You go ahead and investigate 10 different students, how much marks that they have got and you found the average marks are 67, okay? Now because the 67 is way lesser than 85, you now are trying to find out whether the average this year has got lower because over the years, you know the average has been around 85. Now you have investigated 10 different people in your class and you have found their averages are 67.

Now you are trying to find out whether the averages have actually got lesser this year or it is like your samples that you have collected, they were very random and you have collected some samples which out of your entire study. They are just the wide different people you have got, right? So these are the things that we are going to learn in this entire section. I hope this will give you an enough understanding of what we are going to learn.

So this entire process if you think of is 4-step repeatable process and the first step in this case is state the null hypothesis which is H_0 or H_A or state the hypothesis that you have H_0 and H_A . Whatever you are interested in proving, you would keep it in H_A or alternate and everything which does not include alternate is a part of your null hypothesis, okay? So everything minus H_A is your null hypothesis. Whatever you are interested in proving, that is your alternate hypothesis, okay? Then the second thing is set the clear level of significance or level of significance is generally 5%, 1%, 10% that is what you would want to have because level of significance also tells you at what percentage you want to reject or accept claims.

So let's say when you say 5% level of significance, if the p-value comes out to be lesser than 5% or if it comes out to be 0.04, then you will say your alternate hypothesis is true. While if you set the level of significance at 5%, your probability value or p-value comes out to be after this, the entire statistics after the entire calculation, it comes out to be 10%, then you will say that you fail to reject null hypothesis or you accept null hypothesis. Generally, we don't say we accept null hypothesis.

So we say that we fail to reject null hypothesis, okay? Third is we calculate the test statistics. So we calculate the statistic and then using that statistic, either p-test or z-test, if you have large sample size, you can go ahead with z-test. If you have small sample size and the standard deviation is unknown and you use sample standard deviation for that, then you will take t-statistics.



And in that case, if the t-statistics is very large, in that case, you will find that your p-value is lesser and if the p-value is lesser, hence your alternate hypothesis would be true. So this is what your entire situations would be. This is a repeatable process.

First you will state your alternate and null hypothesis. Alternate as I've told you, it is what you are willing to prove, you are willing to justify or what your suspicion is all about. Everything which does not include your alternate hypothesis is a part of your null hypothesis.

Then you set the level of significance. This is required because based on this, you will judge whether your alternate is true or it's false. Then you will calculate the test statistics.

So your test statistics would be based on either z or t. Generally we take if the sample size is good and you have the standard deviation known, then we go ahead with z-test. But that generally is not the case. You do not know the population standard deviation.

In that case, we go ahead with t-test. Remember in last few lectures, we studied the difference between z and t-test. They are exactly same.

It's just that when you do not know population standard deviation, you can go ahead with t-test. So you have t-test and in that case, you will use sample standard deviation as a markup for population standard deviation and you can use a sample t-test. Once you have derived t-test or z-score, t-score or z-score, you can make a decision using p-value.

And this is what we will go through in example later on in this set of slides. So how surprising are our evidences? This brings us to the most important concept in the entire framework, the p-value. Understanding this is key to understanding everything else.

The p-value has a very specific definition. It is the probability of observing data as extreme or even more extreme than what you actually got under the assumption that the null hypothesis is completely true. So the p-value is the probability of getting the sample result as extreme or more extreme than what you have observed assuming that the null hypothesis is always true.

So let's say in your courtroom analogy, this is the answer to the question, right? How likely is this evidence if the defendant is truly innocent? How likely is that evidence provided that the defendant or the null hypothesis is true, defendant is innocent? How likely is the evidence? We are questioning the evidence. What are the chances that the evidence is true provided that the defendant is truly innocent? What is the probability of finding that particular evidence? A small p-value which is 0.01 means



that the evidence is very surprisingly unlikely and this leads to us the doubt. If we have p-value which is very very small, it means that the alternate hypothesis is true.

It means that in this case, the individual is actually a criminal. So if you have a very low p-value, it directly means that the alternate hypothesis is true. If you have a very high p-value, it means that we fail to reject null hypothesis.

That's what has been illustrated here as well. If your p-value is less than or equal to alpha, we generally keep alpha as 5%. If your p-value is less than 5%, then we reject H_0 because we have sufficient evidence.

If p-value is greater than alpha, we fail to reject H_0 because we do not have sufficient evidence in claim of H_1 or alternate hypothesis. So this is really important. One thing that you should take away from this slide is if the p-value is less than 0.5 or if the p-value is lesser than your level of significance 5%, if you want your confidence interval at 95%, it means your level of significance would be 5% and if your p-value in that case is less than 5%, then we reject H_0 or we accept alternate hypothesis.

So that's what it is. If your p-value in that case is greater than alpha, then you fail to reject H_0 . Now, when we do all of these things, we reject H_0 or we fail to reject H_0 , whatever it is, in either of these cases, we can have a situation wherein we can make mistakes.

So there are these two different types of mistakes if you think of with the analogy that we have gone through in this session about the courtroom trial. So first thing is, the defendant is actually innocent and your analysis also says that the defendant is innocent, which is all fine, there is a correct decision. Now if the defendant is actually a criminal and your analysis says that he or she is a criminal, then also that's correct.

But what if the defendant is actually guilty and you let them free. The defendant is guilty but you say that the defendant is not guilty. So in that case, you have letting a guilty person go free and that's what is type 2 error.

The second type of problem is when the defendant is actually innocent and you convict the innocent individual. So you reject H_0 and you accept alternate. When you accept alternate, you are saying that the individual is criminal while the individual in real sense is not a criminal.

So both of these two different types of errors can be type 1 error which is nothing but the level of significance that you assume 5% or 1% or 10%. It means that you are actually rejecting the null hypothesis while performing all the test while in truth, null hypothesis was correct. So you rejected null hypothesis when the null hypothesis was correct.



That is your type 1 error. When you fail to reject null hypothesis, when null hypothesis is false, that is what your type 2 error is. So these are two different scenarios.

If you think of two different types of misclassification would happen or two different types of errors that would happen. If you imagine the courtroom trial, either the individual is guilty, you prove them guilty, that is fine. The individual is not guilty, you prove them not guilty, that is also fine.

These two situations are perfectly correct. But there could be situations when, and this is what generally happens, right, the individuals are actually guilty, but we let them free. So that is type 2 error.

There are situations when the individuals are actually not guilty and we convict them. That's type 1 error. So these are two different situations.

I hope this analogy was good enough for you to understand this entire situation, right. So this is the interactive p-value explorer and this particular thing, let's say your average value is 100. The chart shows the distribution of sample means.

If H_0 is true, when I move the slider, you will see that this entire graph changes. Let's say if I say observed mean is 96 and then so on. So now if you see, when I say observed mean is 96.25, everything is just fine.

The statistics score is, Z-score is minus 1.88 and the two tail p-value is 0.061. Now if you see here, it is 0.06. If I move observed sample mean to still lower, you will see that p-value becomes lesser than 0.05, which is in this case 0.046 and hence you reject null hypothesis. So herein you are saying that because p-value is less than 0.05, you are able to reject null hypothesis or you would accept the alternate hypothesis. Now you see here, on the other side, when your observed mean is this, 104, now also your p-value is less than 0.05 and in that case also your t-statistics too, the two tail p-value is 0.04, which is less than 0.05, hence you reject null hypothesis.

This is the entire simulation drawn just for you to understand what happens, in what scenarios you will reject or accept null hypothesis or you will accept the alternate hypothesis. This is an example. Let us say that a manufacturer claims its phone's battery lasts on an average 40 hours.

The average life of the battery is 40 hours. Now we might have done some studies and we might have found out that the average life is actually less. So our suspicion is the average life is less because that is our suspicion.

I told you that whatever is your suspicion, directly write that suspicion into your alternate. So in this case, your suspicion is the average life is less than 40, while this



is what the claim is. Right? Manufacturer has claimed that the average life is equal to 40.

Now, in this case, you would take the level of significance first. You will define, remember the three, the four-step process, state the hypothesis, state the level of significance. Your level of significance could be 5%, 1%, 10%.

Let's say, let's assume that the level of significance is 5%. Now we are willing to accept 5% chance of typhoon error. Remember that this is what we studied in the last slide.

Now, we test a sample of 9, n is equal to 36 batteries. And we found that the mean of those 36 batteries is 38.8 hours. And now you see that this is lesser than your 40 value, right? Now let's assume that the standard deviation is also 3. So what we do, we calculate Z statistics, which is nothing but the computed test statistic from the sample and your actual population value, the difference between them.

So this is sample mean minus population mean divided by sigma by root n , which is the standard error. Now, once you do this, once you calculate this, you feed in all the values that you have got, you are able to find that Z is equal to minus 2.40. Now, add this Z value, we will find out the P value and P value comes out to be 0.082 because the P value is less than 0.05. And because your level of significance was 5%, hence your P value is less than your level of significance, which means that you would reject the null hypothesis and you will accept alternate. So this is how you generally do your hypothesis testing.

These are the scenarios that you will have to assume. I hope this was all clear with everyone. In summary, the hypothesis testing framework gives us a disciplined process to evaluate claims with evidence.

In the next video, we will look at how errors, P values, significance levels guide our decisions.

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